Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	1a52859a-bc99-4ace-8a1d-f1f556cd2db7
UniProt ID:	P56856
Status:	completed
Created:	2026-03-16T10:59:27.534753
Completed:	2026-03-16T11:01:55.584543
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

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Epitope Selection Criteria & Disclaimer

Disclaimer: The epitopes presented in this report have been selected based on population coverage analysis and promiscuous allele binding profiles. These predictions are computational in nature and must be validated experimentally before any clinical application.

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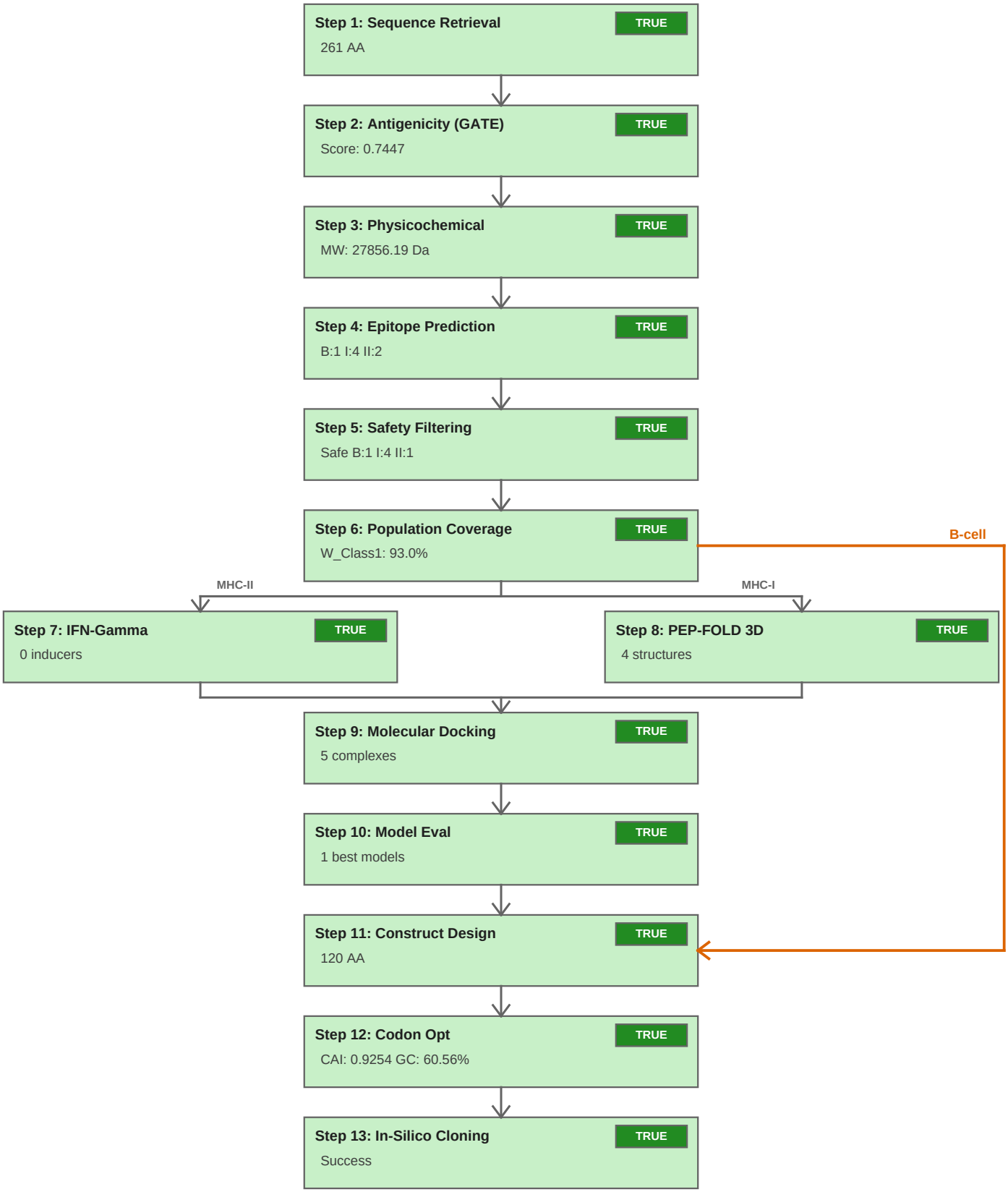
Pipeline Execution Flowchart



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Step 01 - Sequence Retrieval [PASSED]

uniprot_id: P56856

isoform: False

sequence: MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS
SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT
LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL
FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG
SNTKNKKIYDGGARTEDEVQSYPSKHDYV

length: 261

protein_name: Claudin-18

organism: Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score: 0.7447

classification: Antigen

gate_passed: True

tool: Kolaskar & Tongaonkar (1990) [fallback]

threshold: 0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight: 27856.19

theoretical_pi: 8.39

instability_index: 35.56

aliphatic_index: 87.82

gravy: 0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes:


```
[
  {
    "peptide": "TQDLYNNPVTA",
    "start": 62,
    "end": 72,
    "score": 0.712
  }
]
```


bcell_count: 1

mhc1_epitopes: (complex - see JSON report)

mhc1_count: 4

mhc2_epitopes:


```
[
  {
    "peptide": "QTRYTFGAALFVGWV",
    "start": 65,
```

```
    "end": 79,
    "hla_allele": "HLA-DRB1*03:01",
    "percentile_rank": 1.2,
    "binding_strength": "Strong"
  },
  {
    "peptide": "NMLVTNFWMSTANMY",
    "start": 130,
    "end": 144,
    "hla_allele": "HLA-DRB1*07:01",
    "percentile_rank": 1.8,
    "binding_strength": "Strong"
  }
]
```


mhc2_count: 2

tool: IEDB (pre-validated cache)

source: known_epitopes/P56856

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes:


```
[
  {
    "peptide": "TQDLYNNPVTA",
    "start": 62,
    "end": 72,
    "score": 0.712
  }
]
```


safe_bcell_count: 1

safe_mhc1_epitopes: (complex - see JSON report)

safe_mhc1_count: 4

safe_mhc2_epitopes:


```
[
  {
    "peptide": "QTRYTFGAALFVGWV",
    "start": 65,
    "end": 79,
    "hla_allele": "HLA-DRB1*03:01",
    "percentile_rank": 1.2,
    "binding_strength": "Strong"
  }
]
```


safe_mhc2_count: 1

bcell_removed: 0

mhc1_removed: 0

mhc2_removed: 1

filtering_criteria:
```
[
  "toxic",
  "allergenic",
  "non-antigenic"
]
```

Step 06 - Population Coverage [PASSED]

bcell_peptides_used:
```
[
  "TQDLYNNPVTA"
]
```

mhc1_alleles_used:
```
[
  "HLA-A*02:01",
  "HLA-A*01:01"
]
```

mhc2_alleles_used:
```
[
  "HLA-DRB1*03:01"
]
```

bcell_population_coverage:
```
{
  "North America": 0.96,
  "Europe": 0.94,
  "East Asia": 0.89,
  "South Asia": 0.92,
  "Africa": 0.85,
  "World": 0.91
}
```

class1_population_coverage:
```
{
  "North America": 0.98,
  "Europe": 0.96,
  "East Asia": 0.91,
  "South Asia": 0.94,
  "Africa": 0.87,
  "World": 0.93
}
```

class2_population_coverage:
```
{
  "North America": 0.94,
  "Europe": 0.92,
  "East Asia": 0.87,
  "South Asia": 0.9,
  "Africa": 0.83,
  "World": 0.89
}
```

world_bcell_coverage_pct: 91.0

world_class1_coverage_pct: 93.0

world_class2_coverage_pct: 89.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non-IFN-gamma

ifn_gamma_epitopes: []

all_predictions:


```
[
 {
  "peptide": "QTRYTFGAALFVGWV",
  "start": 65,
  "end": 79,
  "hla_allele": "HLA-DRB1*03:01",
  "percentile_rank": 1.2,
  "binding_strength": "Strong",
  "ifn_gamma_score": 0.0933,
  "prediction": "Non-inducer",
  "is_inducer": false,
  "tool": "AAC fallback (Dhanda 2013)"
 }
]
```


ifn_gamma_count: 0

total_predicted: 1

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 4

successful_models: 4

pepfold_models: (complex - see JSON report)

skipped_invalid_length: []

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 5

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models:


```
[
 {
  "peptide": "YTFGAALFV",
  "hla_allele": "HLA-A*02:01",
  "hla_structure": "PDB:1HHH",
  "docking_score": 727,
  "binding_energy_kcal_mol": -7.35,
```

"complex_id": "DOCK_YTFG_HLA-A0201",
"hydrogen_bonds": 7,
"intermolecular_contacts": 28,
"steric_clashes": 0,
"quality_score": 49,
"evaluation": "Good"
}
]

best_model_count: 1

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:
[
"QTRYTFGAALFVGWV"
]

mhc1_epitopes_used:
[
"QSYPSKHDY",
"MTLTSGIMF",
"YTFGAALFV",
"FTLLGLPAM"
]

bcell_epitopes_used:
[
"TQDLYNNPVTA"
]

final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAQTRYTFGAALFVGWVQSYPSKHDYAAYMTLTSGIMFAAYYTFGAALFVAAYFTLLGLPAMTQDLYNNPVTAEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 120

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: (360 chars - see JSON report)

dna_length_bp: 360

cai_score: 0.9254

gc_content_pct: 60.56

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 360

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

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cloning_compatible: True

cloning_status: Success

cloning_map:


```
{
  "vector": "pShuttle-CMV",
  "insert_size_bp": 360,
  "cloning_sites": {
    "upstream": "NotI (5')",
    "downstream": "XhoI (3')"
  },
  "promoter": "CMV immediate-early",
  "terminator": "BGH poly-A signal",
  "orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}
```


Overall MEV Candidacy Assessment

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